

Antonio Barone Nassar

**OLI (Optimized Lead Intelligence): An AI-Native Multi-Channel
Virtual Receptionist for Small Service Businesses in Brazil**

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Antonio Barone Nassar

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Final Course Project submitted to the Institute of Technology and Leadership (INTELI), to obtain a bachelor's degree in Software Engineering.

Advisors:

Prof. Thais Rodrigues Neubauer (Modules 13 and 14)

Prof. José Romualdo Costa Filho (Modules 15 and 16)

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RESUMO

NASSAR, Antonio Barone. OLI (Optimized Lead Intelligence): um recepcionista virtual multicanal com inteligência artificial para pequenos negócios de serviços no Brasil. 2026. n° de folhas. TCC (Graduação), Curso de Engenharia de Software, Instituto de Tecnologia e Liderança, São Paulo, 2026.

Pequenos negócios de serviços no Brasil, como salões de beleza, clínicas de estética, pet shops e estabelecimentos de bem-estar, operam majoritariamente por aplicativos de mensagens, sobretudo o WhatsApp, mas perdem clientes quando ninguém consegue responder a tempo, além de sofrerem com agendamentos manuais e taxas de falta que chegam a 20 a 30 por cento. Este trabalho apresenta o OLI (Optimized Lead Intelligence), uma plataforma B2B nativa de inteligência artificial que atua como recepcionista virtual nos canais de mensagens do negócio, principalmente o WhatsApp, mas com arquitetura agnóstica a canal (Telegram e Instagram Direct): lê as mensagens, compreende a intenção, responde em português, agenda atendimentos e envia lembretes, enquanto um painel web multiempresa permite ao dono acompanhar conversas, gerenciar agenda e serviços, configurar o agente e assumir o atendimento quando necessário. O projeto foi conduzido ao longo de quatro módulos acadêmicos: pesquisa e personas (M13), desenvolvimento do MVP (M14), avaliação de product-market fit (M15) e consolidação da plataforma, modelo de negócio e apresentação final (M16). A validação inicial, em pré-piloto com três empresas reais, associou o valor percebido à rapidez da primeira resposta e à redução de leads perdidos, condicionados a um onboarding simples, a controle visível e à ancoragem das respostas no catálogo de serviços. Conclui-se que a abordagem centrada em mensagens, com um modelo de negócio viável baseado em assinatura, constitui uma base sólida e bem documentada para um negócio que atende a uma necessidade real de um mercado amplo e pouco servido.

Palavras-Chave: Inteligência Artificial; Comunicação por Aplicativos de Mensagens; Automação para Pequenas Empresas; Agendamento; Product-Market Fit; Plataforma SaaS.

ABSTRACT

NASSAR, Antonio Barone. OLI (Optimized Lead Intelligence): An AI-Native Multi-Channel Virtual Receptionist for Small Service Businesses in Brazil. 2026. n° of pages. Final course project (Bachelor), Course Software Engineering, Institute of Technology and Leadership, São Paulo, 2026.

Small service businesses in Brazil, such as beauty salons, aesthetic clinics, pet grooming shops, and wellness practices, operate mostly through messaging apps, above all WhatsApp, yet they lose clients when nobody can reply in time, and they struggle with manual scheduling and no-show rates of twenty to thirty percent. This work presents OLI (Optimized Lead Intelligence), an AI-native B2B platform that acts as a virtual receptionist on the business's messaging channels, primarily WhatsApp but with a channel-agnostic architecture that also covers Telegram and Instagram direct messages: it reads incoming messages, understands intent, replies in Portuguese, books appointments, and sends reminders, while a multi-tenant web dashboard lets the owner follow conversations, manage the agenda and service catalog, configure the assistant, and take over when a human is needed. The project was developed across four academic modules: research and personas (M13), MVP development (M14), product-market fit assessment (M15), and platform consolidation, business model, and final presentation (M16). Early validation, in a pre-pilot with three real businesses, linked perceived value to the speed of the first reply and to fewer missed leads, conditioned on simpler onboarding, visible control, and answers grounded in the service catalog. The study concludes that a messaging-first approach, supported by a viable subscription-based business model, provides a solid and well-documented foundation for a venture that addresses a genuine need in a large and underserved market.

Key words: Artificial Intelligence; Conversational Messaging; Small-Business Automation; Appointment Scheduling; Product-Market Fit; SaaS Platform.

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LIST OF ABBREVIATIONS AND ACRONYMS

AI Artificial Intelligence

API Application Programming Interface

ARR Annual Recurring Revenue

B2B Business-to-Business

CRM Customer Relationship Management

LGPD Lei Geral de Proteção de Dados (Brazilian General Data Protection Law)

LLM Large Language Model

MAU Monthly Active Users

MVP Minimum Viable Product

PMF Product-Market Fit

RLS Row-Level Security

SaaS Software as a Service

SAM Serviceable Available Market

SMB Small and Medium-sized Business

SOM Serviceable Obtainable Market

TAM Total Addressable Market

WAHA WhatsApp HTTP API

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1 INTRODUCTION

Small service businesses are the backbone of everyday commerce in Brazil, and for most of them messaging apps, and WhatsApp in particular, are the operational hub through which clients ask about services, check prices, and try to book appointments. Roughly eighty percent of bookings in Brazil start on WhatsApp (Meta, 2024), and ninety-eight percent of WhatsApp messages are opened, against about twenty percent of emails (Opinion Box, 2026). The channel already exists; what is missing is a way to answer fast and consistently.

This creates a structural problem. When the owner is serving a client, at lunch, or off-hours, messages sit unanswered, and clients who do not hear back quickly often move to a competitor. The result is a steady, mostly invisible loss of revenue. The pain was first observed up close inside a corporate sales team, where the volume of unanswered messages was large, leads arrived through WhatsApp with no coverage outside business hours, and the cost was constant but invisible. The founding hypothesis of this project is that the same pain applies, at much larger scale, to small and medium-sized businesses such as clinics, hair salons, and pet shops, which receive bookings through messaging but lack the scale or budget for a corporate solution.

This Final Course Project presents OLI (Optimized Lead Intelligence), an AI-native business-to-business platform that addresses this gap. OLI works as a virtual receptionist on the business's own messaging channels, primarily WhatsApp but with a channel-agnostic design that also covers Telegram and Instagram direct messages: it reads incoming messages, understands what the client wants, replies in a natural and configurable tone, books appointments directly into the business's calendar, and sends reminders, while the owner always retains control through a multi-tenant web dashboard. The guiding design principle is to meet businesses where they already are, augmenting their existing workflow rather than forcing them to migrate to a new system.

The project was not conceived as a finished product from the outset. It began as a research question and matured across four academic modules, each answering a distinct question: whether a real problem worth solving exists (Module 13), whether it can actually be built (Module 14), whether real customers find it valuable (Module 15), and whether it can be presented as a coherent, economically sustainable venture (Module 16).

1.1 Objectives

The general objective of this work is to design, build, and validate an AI-native, multi-channel virtual receptionist that automates the first response and appointment scheduling for small service businesses in Brazil. The specific objectives are:

- characterize the operational problem and quantify the market through field interviews and secondary data;
- build a multi-tenant platform combining a conversational messaging agent and a management dashboard, integrated with calendar and scheduling;
- formulate testable product-market fit hypotheses and assess them through validation with real businesses;
- specify a sustainable business model, with pricing and unit economics consistent with the target market.

2 DEVELOPMENT

This section follows the venture from problem discovery to a consolidated, demo-ready platform. It describes the market research that defined the problem (2.1); the size and structure of the market (2.2); the solution and the product as built (2.3); the system architecture (2.4); the methodology and hypotheses used to validate it (2.5); the results of that validation (2.6); and a discussion that integrates the evidence with the business model and the project's limitations (2.7).

2.1 Discovery and Problem Understanding

The first module of the venture was dedicated entirely to understanding the market before writing a single line of product code. Over a research cycle that combined field interviews, competitive analysis, persona development, and pain point mapping, the goal was to determine whether the perceived problem was real, widespread, and worth building for, letting that evidence, rather than assumptions, shape every later decision.

Three in-depth interviews were conducted across different segments. An aesthetics and dermatology clinic had been using clinic software for over two years but still scheduled appointments entirely by hand through messaging apps, with staff juggling a CRM, spreadsheets, and paper records; the owner wanted a messaging assistant for pre-triage and appointment confirmation, targeting a thirty percent reduction in no-shows and forty percent faster confirmations. A beauty salon, operating a salon-focused system for more than twenty years, also scheduled fully by hand and reported no-show rates of thirty to forty percent for

manicures, while tracking over two hundred hair-color products manually. A media production house, although less centered on messaging, struggled with manual file handling and the absence of predictive insight.

Across all three conversations a consistent pattern emerged: messaging apps, and WhatsApp above all, are the operational hub for client communication; records are fragmented; scheduling is manual and error-prone; no-shows waste capacity; and existing software is rigid, poorly integrated, or too expensive. A competitive analysis confirmed that most available tools cover basic scheduling but lack messaging-first integration, affordable AI automation, simplicity for small teams, and compliance designed for Brazilian regulations such as the LGPD. From these insights, three customer personas were defined: health-and-beauty service providers, neighborhood small-business owners, and creative or professional enterprises. The strongest product-market fit signal came from the first two profiles, namely businesses where appointments drive revenue and messaging is the primary client channel. Table 1 summarizes the venture journey that this research initiated.

Table 1. The four-module venture journey and the question each answered.

Module	Focus	Central question	Outcome
13	Market research and validation	Is there a real problem worth solving?	Evidence from interviews, competitive analysis, personas, and MVP scope.
14	Entrepreneurship and MVP development	Can we actually build this?	A working messaging assistant and a bilingual landing page.
15	Product development and PMF	Do real customers find this valuable?	Multi-tenant design and a three-business pre-pilot with qualitative evidence.
16	Consolidation, pitch, and business model	Can we present this as a coherent venture?	Unified oli-app platform, MAU business model, and a board presentation.

2.2 Market and Opportunity

The problem identified in the field is backed by the size and behavior of the Brazilian market. There are roughly 1.5 million appointment-based service businesses in Brazil (SEBRAE, 2025), and about seventy-two percent of small businesses already use WhatsApp to serve customers (RD Station, 2024), within a base of 147 million active WhatsApp users that corresponds to 93.4 percent of the online population (Sinch, 2024). Messages on the channel are almost always read, far more than email (Opinion Box, 2026), yet most businesses still respond

manually and have no coverage outside business hours.

Two consequences make the problem expensive. No-show rates of twenty to thirty percent are common in clinics and salons, and automated reminders can reduce them by up to sixty percent (ByDoctor, 2026); and, since around eighty percent of bookings in Brazil start on WhatsApp (Meta, 2024), a fast automated first response acts directly on revenue. As a reference for value, the true cost of a full-time receptionist reaches between R\$ 1,670 and R\$ 2,450 per month once Brazilian labour charges are included (CAGED and Portal Salário, 2026).

Translating this into a market sizing, the total addressable market is estimated at around R\$ 3.1 billion per year, the serviceable available market at around R\$ 470 million per year (businesses with an active WhatsApp Business profile), and a realistic first-year serviceable obtainable market at about R\$ 84 thousand in annual recurring revenue under a founder-led operation in São Paulo. Table 2 summarizes these figures.

Table 2. Market opportunity (TAM, SAM, SOM).

Layer	Estimate	Basis
TAM	R\$ 3.1 billion / year	~1.5M appointment-based businesses, average ticket R\$ 174/month (SEBRAE, 2025).
SAM	R\$ 470 million / year	~225k businesses with active WhatsApp Business (~15% of TAM).
SOM	R\$ 84 thousand / year (ARR)	Year-1 target of 40 paying businesses, founder-led, São Paulo.

The competitive landscape is fragmented. Vertical scheduling systems such as Booksy, Trinks, and Clínica nas Nuvens cover bookings but are not messaging-first and offer little conversational automation, while message-automation tools such as ManyChat rely on static flows rather than a true conversational agent. Closer competitors include Atendente.ai, a generic Brazilian WhatsApp tool, and Wazzy.io, a European WhatsApp product. OLI is positioned for the Brazilian SMB segment with three differentiators: a genuinely conversational AI agent rather than a menu or static flow, a channel-agnostic design instead of a single-channel integration, and a clear MAU-based price with a low entry point.

2.3 Solution and Product

The second module turned research into a working proof of concept, and the following modules grew it into a complete product. At its core, OLI is an AI agent

that serves customers around the clock in the channel they already use, books directly into Google Calendar, and sends automatic reminders to reduce no-shows. A client can open a conversation, ask for an appointment, receive available slots, confirm, and get a calendar confirmation, all without human intervention.

Around the agent sits an interactive dashboard for management, metrics, and configuration. The owner can view the schedule and bookings in real time, configure the agent (business hours, services, and custom responses), and follow business metrics such as appointments, occupancy rate, and agent performance. The delivered feature set, frozen for the board presentation, includes a conversational agent with seven tools, autonomous booking, rescheduling and cancellation, real-time Google Calendar integration, a dashboard with live conversations and schedule, a lightweight CRM for clients, professionals, and services, and human handoff, both manual and automatic.

2.4 System Architecture and Platform Consolidation

The most significant technical evolution of the project occurred in the final module, when the earlier workflow-based prototype, assembled with visual automation tooling, was migrated into a single, full-code application named oli-app. The migration was motivated by practical needs: serving multiple businesses on one platform required an architecture that visual tools could not support cleanly; a coded application is easier to maintain; and modern AI development tooling works better with structured code than with visual flow diagrams. The stack is built on Next.js 14 with shadcn/ui, the Vercel AI SDK, Supabase, and integrations with Google Calendar, Resend for email, and Asaas for billing. Figure 1 presents the system context and how the components communicate through their APIs, and Figure 2 the deployment view.

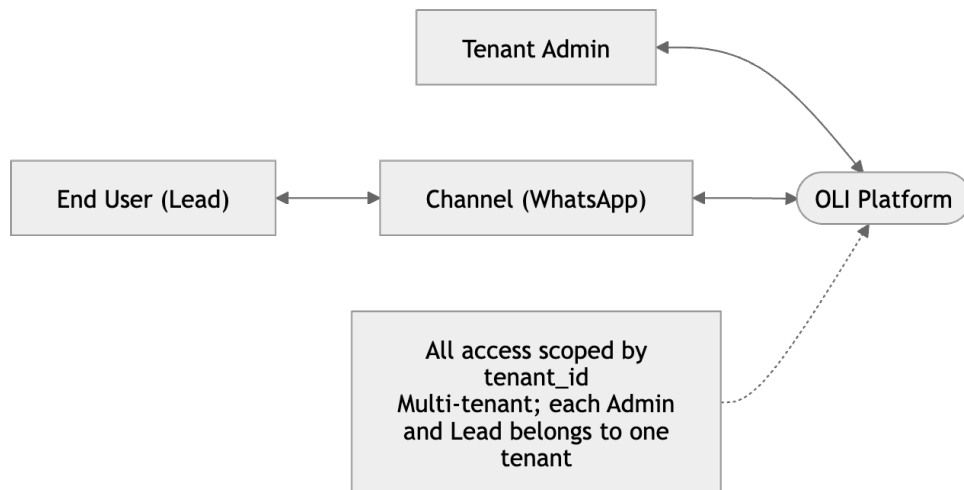


Figure 1. OLI system context diagram (component and API communication).

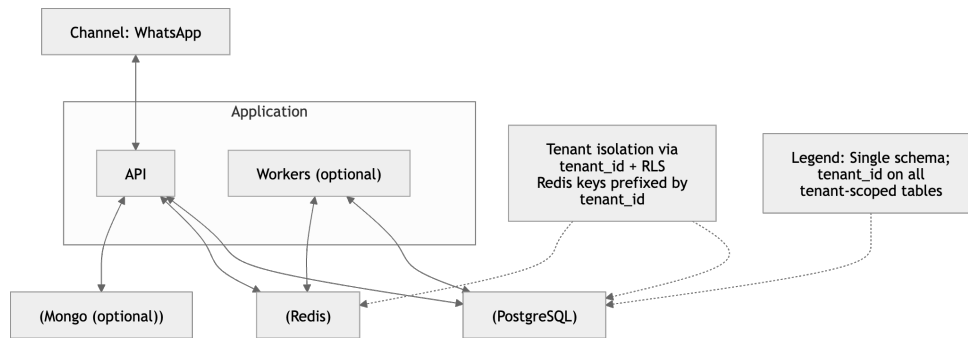


Figure 2. Deployment view of the consolidated oli-app stack.

The consolidated platform is a single Next.js application (App Router) that hosts both the web dashboard and the AI agent surface, backed by Supabase (PostgreSQL) in production and staging environments. Strict separation between tenants is enforced through Row-Level Security, so that each business sees only its own data. The remainder of this section describes the main modules.

2.4.1 Channels and Messaging Layer

Entry channels are abstracted behind a messaging layer so that the rest of the system is independent of any single provider. WhatsApp is the first and primary channel, connected through WAHA (a self-hosted WhatsApp HTTP API) running as a Docker container suitable for deployment on a virtual private server; Telegram is supported through the Telegram Bot API, and Instagram direct messages are part of the same channel-agnostic design. Incoming messages flow through a webhook into the agent pipeline, and replies return through the originating channel. A migration to the official WhatsApp Business API, through Meta, was in progress at the time of writing.

2.4.2 Front-End and Dashboard

The dashboard is the surface through which the business owner and their team operate the product. It provides a home overview of the day's appointments and active conversations; an agenda integrated with Google Calendar and filterable by professional; a real-time conversations view with transfer to human, which pauses the assistant and notifies the team; client records with conversation history, notes, and tags; a professionals area where each team member can connect their own calendar; a services catalog that the assistant uses as its single source of truth for prices; agent settings for name, tone, instructions, business hours, and manual-confirmation behavior; business metrics; and user management for operators and administrators.

2.4.3 Back-End and Data Layer

The back-end is implemented within the same Next.js project through server actions and API routes. Authentication is handled by Supabase Auth, with session cookies and route protection through middleware; on sign-up, a bootstrap routine provisions the business workspace. The data layer is a multi-tenant PostgreSQL schema covering businesses, users, professionals and their calendar tokens, services, clients, conversations, messages, appointments, and agent settings, with TypeScript types generated directly from the schema and Row-Level Security isolating each tenant.

2.4.4 AI Agent

The conversational intelligence is orchestrated with the Vercel AI SDK over a large language model, configured per business. Rather than producing free-form answers, the agent is equipped with seven tools backed by the calendar and the database: `check_availability`, `create_appointment`, `cancel_appointment`, `list_appointments`, `send_reminder`, `get_business_info`, and `handle_faq`. Because the agent acts through these tools, its responses are grounded in the business's real data: it quotes only prices that exist in the catalog and schedules only against real availability, directly addressing the accuracy concerns raised by pre-pilot participants.

2.4.5 Integrations

Beyond the messaging channels, three integrations complete the platform. Google Calendar is connected through OAuth with per-professional token refresh, enabling the agent to read availability and create appointments in real time. Resend delivers email confirmations to clients and professionals when

appointments are booked or when a conversation is escalated to a human. Asaas provides billing through PIX, bank slip, and card, supporting the subscription model described in the discussion.

2.5 Methodology and Validation

To assess whether the product delivers real value, the project followed a product-market fit approach rather than a controlled laboratory experiment, in line with build-measure-learn practice (Ries, 2011). The design is qualitative and field-based, centered on a small number of real businesses using the platform in their day-to-day operations, because the research questions concern perceived value, trust, and adoption, which are best captured through direct observation and interviews with the people who would actually pay for and operate the product.

2.5.1 Hypotheses

Six testable hypotheses were defined, each with a metric and a success threshold, to structure the validation and to support a later decision to persevere, pivot, or adjust segment and pricing. Table 3 summarizes them.

Table 3. Testable product-market fit hypotheses and success thresholds.

#	Hypothesis focus	Success threshold
H1	Value (response time and usefulness)	≥ 50% improve first-response time; ≥ 80% agree it helps respond faster.
H2	Willingness to pay	≥ 60% would pay at or near the target price.
H3	Activation (time to value)	≥ 80% activate within 2 weeks of go-live.
H4	Retention (continued use)	≥ 80% active use; zero unplanned churn.
H5	Lead quality and conversion	≥ 1 lead per business per week; ≥ 70% agree it helps capture leads.
H6	Onboarding feasibility	Median go-live ≤ 5 business days; ≤ 4 hours of support per business.

2.5.2 Participants and Procedures

Three businesses took part in the pre-pilot, each within the priority segments identified during research: one aesthetics clinic, one hair salon, and one pet-services business, all of which expressed interest in the solution as paying customers. Participants are identified by role and segment only, to protect their privacy. The planned cadence comprised an onboarding conversation to align expectations and connect the messaging channel, a mid-pilot check-in to capture early friction, and a closing interview on value, pricing, and improvements,

complemented by a short satisfaction survey. Qualitative data were examined thematically, grouping observations into recurring themes (value, onboarding, AI accuracy and control, trust, and channel-policy concerns), and improvement requests were prioritized by expected impact and implementation effort.

2.6 Results

2.6.1 Qualitative Findings

Across the three businesses, participants consistently associated the platform's value with not losing leads after hours and with faster first replies when the team is busy; several described the agent as a useful first filter so that staff handle only complex cases. Each business surfaced a concrete, actionable need: the aesthetics clinic asked for importing its client base from a legacy CRM, the hair salon described the platform as simple and easy to use, and the pet-services business asked for routing a conversation to a specific staff member. On accuracy and trust, owners valued configuring the agent's name and tone but asked for an easy way to pause the AI for a specific conversation; answers were trusted when prices and services matched the configured catalog, and mismatches caused concern, reinforcing the importance of catalog grounding. Reliance on a single messaging provider and the risk of policy changes or bans remained an explicit concern across the three businesses.

2.6.2 Assessment Against Hypotheses

Because the validation was an early pre-pilot within an academic timeline, the hypotheses received qualitative rather than fully instrumented support. The value hypothesis (H1) and the willingness-to-pay hypothesis (H2) found clear supportive signal in interviews, with owners explicitly valuing faster first replies and comparing the price favorably to part-time staff. Onboarding feasibility (H6) looked achievable within about a week when a single engaged contact owned the process. The activation, retention, and lead-conversion hypotheses (H3, H4, H5) could not be confirmed quantitatively, because the corresponding instrumentation (first-response time, weekly active use, and lead events per tenant) was only partially in place; confirming them against the defined thresholds is the central task of a longer, fully instrumented pilot.

2.6.3 Prioritized Improvements

Synthesizing the feedback produced a prioritized list of improvements, shown in Table 4. The recommended sequencing ships stricter catalog grounding and onboarding self-help first, both low-effort and high-value risk reducers, followed

by priority-lead notifications and per-conversation human takeover once the backend conversation state is defined.

Table 4. Prioritized pre-pilot improvements (impact and effort).

#	Improvement	Impact (1 to 5)	Effort (1 to 5)
1	Notification for new priority leads	5	3
2	“Pause AI” / human takeover per conversation	5	4
3	Strict grounding: block prices not in catalog	4	2
4	In-dashboard checklist for channel reconnect	4	2
5	Import client base from legacy CRM / export contacts	3	3

2.7 Discussion

2.7.1 Product-Market Fit Signals

The evidence answers the central question of the validation phase: real customers do find the product valuable, but under clear conditions. The value proposition that resonated most was not a long feature list but a single, concrete benefit: the speed of the first reply and the resulting reduction in missed leads, especially after hours and on weekends. Three conditions consistently shaped whether that value was realized: onboarding must be simple, control must be visible (pausing the AI, transferring to a human, reading full threads), and answers must stay grounded in what the business actually offers. These conditions, rather than additional capabilities, define the product’s near-term roadmap.

2.7.2 Business Model

OLI is offered as a business-to-business software-as-a-service product through a monthly subscription. Pricing is based on Monthly Active Users (MAU), the number of unique clients who contact the business through its messaging channels in a given month; this metric was adopted, after mentoring feedback, in place of an earlier and more ambiguous “conversations per month,” because it is understandable to owners and scales fairly with the value delivered. Three tiers are offered, summarized in Table 5; billing runs through Asaas with PIX, bank slip, or card, and can be cancelled at any time. Positioned against the cost of a full-time receptionist, the product is an affordable alternative to expanding front-desk capacity.

Table 5. Subscription tiers (MAU-based).

Tier	Price / month	MAU	Highlights
Essential	R\$ 147	Up to 100	1 professional, Google Calendar, AI agent and dashboard.
Professional	R\$ 247	Up to 300	Up to 5 professionals, CRM and human handoff, advanced metrics.
Business	R\$ 447	Up to 600	Unlimited professionals, advanced agent config, priority support.

The internal cost structure and unit economics behind these tiers were developed during the venture and presented to the evaluation board; as company-specific financial information, they are intentionally kept out of this public report.

2.7.3 Limitations

A credible account must state what the project did not achieve. The validation sample was small, three businesses in an early pre-pilot, which provides deep qualitative insight but cannot statistically represent the broader market, and several hypotheses remain to be confirmed with full instrumentation. The work was conducted within an academic program with fixed timelines rather than as a commercial startup. The platform currently relies on WhatsApp as its primary channel, so changes in Meta's policies remain a real risk until additional channels are fully exercised, and the official WhatsApp Business API approval was still in progress. At the time of the board presentation the platform was fully functional in development and staging, but a complete production deployment pipeline and self-serve onboarding were planned and not yet finished. Analytics instrumentation was partial, and some requested features, such as per-conversation pause and operator routing, remained on the roadmap.

3 FUTURE WORKS

Several directions follow naturally from the evidence gathered. On the product side, the priorities are completing a stable production deployment with the official WhatsApp Business API and a self-serve onboarding flow; implementing per-conversation pausing of the AI and routing to a specific professional; enforcing strict catalog grounding so the agent never quotes prices outside the configured services; adding notifications for new priority leads and AI escalations; importing a client base from legacy CRMs; and exercising the additional messaging channels (Telegram and Instagram) already supported by the architecture.

On the business side, the next steps follow the venture roadmap: a go-to-market phase bringing the first paying clients onto the platform, an expansion phase adding integrations with the systems these businesses already use, and longer-term growth driven by referrals and content. The subscription approach should continue to be validated with customers as the base grows.

On the technical side, future work includes completing an end-to-end verification of tenant data isolation, building automated monitoring for message delivery health, and instrumenting the metrics needed to confirm the activation, retention, and lead-conversion hypotheses against their thresholds. These are directions rather than commitments, but together they describe a realistic continuation of the venture beyond the academic track.

4 CONCLUSION

OLI started as an idea: that small Brazilian service businesses, which run on messaging apps but lose clients when nobody answers in time, could be helped by an AI assistant that replies and books appointments for them. Across four modules, that idea was refined step by step. Market research confirmed the problem was real and the market large; the first version proved the assistant could be built and hold real conversations in Portuguese; validation with real businesses showed the value was perceived; and the final stage consolidated everything into a single, working multi-channel platform with a defined subscription business model.

The project is concluded and a venture was created. The idea was developed and validated, the product was built and shown to work, the business model was defined, and real businesses engaged as potential paying customers. OLI is therefore an operating service, ready to continue growing from the foundation built here.

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